

Prediction of Supply Chain Health Using Satellite Imagery, Port Operations Data, and Media Trends: A Multi-Source Analysis of the Port of Los Angeles

Abstract

This paper presents a multi-source data fusion approach for predicting supply chain health by analyzing the Port of Los Angeles using high-resolution satellite imagery, real-time vessel operations data, and global media coverage. A convolutional neural network (CNN) ensemble was applied to PlanetScope satellite images to detect port activity patterns over time. Port data, including vessel counts and turnaround times, were visually analyzed to validate congestion patterns. Media coverage statistics were extracted using News API to assess the public and industry awareness of logistical disruptions. Results indicate strong alignment between satellite-detected port activity, operational port metrics, and spikes in news article frequencies during the 2021 global supply chain crisis. The research confirms that integrating satellite vision with ground truth and social signals provides a robust framework for monitoring and predicting supply chain resilience.

Keywords

Supply chain prediction, satellite imagery, port congestion, CNN, PlanetScope, Port of Los Angeles, logistics, vessel turnaround, news analysis

I. Introduction

In recent years, disruptions in global supply chains have drawn increasing attention, particularly during crises such as the COVID-19 pandemic and the Suez Canal blockage in 2021. Ports serve as critical nodes in global logistics networks, and their operational efficiency directly affects global trade. The Port of Los Angeles, being the busiest container port in the United States, became a focal point for congestion and shipping delays during this period.

This study explores a novel, multi-source approach for predicting supply chain health by treating three independent data streams—satellite imagery, vessel operations data, and media coverage—as corroborating indicators. We aim to evaluate how these streams can collectively inform real-time or near-real-time assessment of port congestion and broader logistical stress.

II. Data Sources and Collection

A. Satellite Imagery: PlanetScope Constellation

High-resolution satellite images of the Port of Los Angeles were obtained from Planet.com via the Planet Explorer platform. PlanetScope imagery offers approximately 3-meter-per-pixel resolution, enabling visibility of ships, container stacks, and port equipment.

- **Area of Interest (AOI):** Terminal Island and surrounding harbor
- **Coordinates:** ~33.74805°N, 118.10055°W

- **Observation Period:** August 2021 and 2023
- **Platform Used:** [Planet Explorer](#)

Cloud cover and image availability varied across dates. Detection counts were extracted daily using a CNN-based detection pipeline.

B. Port Operations Data

Vessel movement and turnaround data were extracted from the Port Optimizer Dashboard ([tower.portoptimizer.com](#)). Key datasets used include:

- **Vessel Count by Month (2019–2024)**
- **Average Turnaround Time (2019–2024)**

These metrics served as ground truth indicators of port congestion during key time periods.

C. Media Coverage Data

A global news database was accessed through [NewsAPI.ai](#) to assess frequency of logistics-related keywords. Datasets included:

- **Article Counts by Keyword (2019–2024)**
- Keywords analyzed: *port congestion, shipping delays, container shortage, labor shortage, Suez Canal blockage*, among others.

III. Methodology

A. Image Analysis via CNN Ensemble

A deep learning-based detection pipeline was adapted from the open-source repository [maups/covid19-satellite-analysis](#). The model utilizes:

- **CNN Ensemble**
Lightweight convolutional neural networks (CNNs) trained on satellite imagery to recognize port activity markers.
- **Strategic Location Sampling**
Grid sampling of satellite images within AOI to optimize detection of ships and equipment.
- **Detection Outputs**
Daily counts of identifiable port-related objects (e.g., ships, containers) used to measure operational intensity.

Detection trends from August 2021 (congestion period) were compared with the same window in 2023 (baseline period).

B. Port Data Visualization

No ML modeling was applied to port data; instead, visual analysis and statistical plots were used to validate congestion patterns. Observations included:

- **Sustained Vessel Volume in 2021:** Monthly vessel counts stayed high with no seasonal dip.
- **Elevated Turnaround Times:** Averaging 85–95 hours in mid-2021, indicating processing delays.

C. Media Signal Analysis

Keyword frequency analysis from 2019 to 2021 revealed:

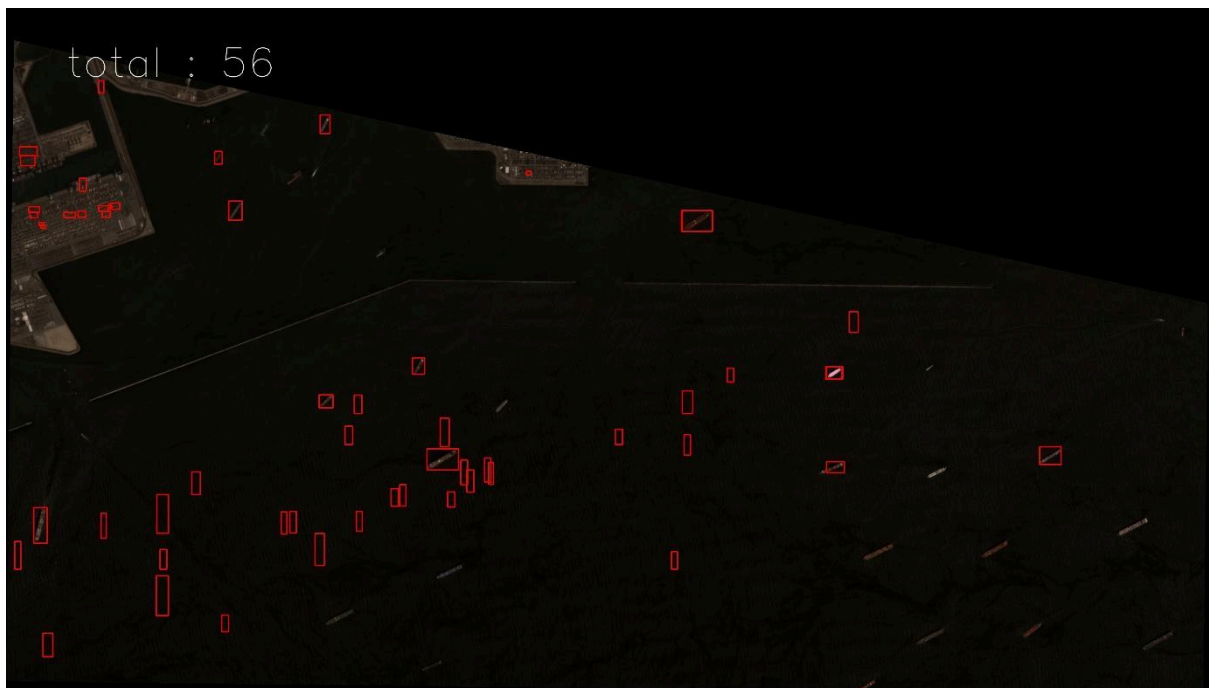
- **“Port Congestion”:** Grew from 562 mentions in 2019 to 6,487 in 2021 (Q4 peak: 2,813).
- **“Shipping Delays”:** From 473 (2019) to 12,153 (2021); Q4 alone had 6,808 articles.
- **“Labor Shortage”:** Jumped to 28,024 articles in 2021, showing systemic strain.
- **Emergent Keywords:** *rotterdam port strike*, *shanghai port delay* emerged only in 2021.

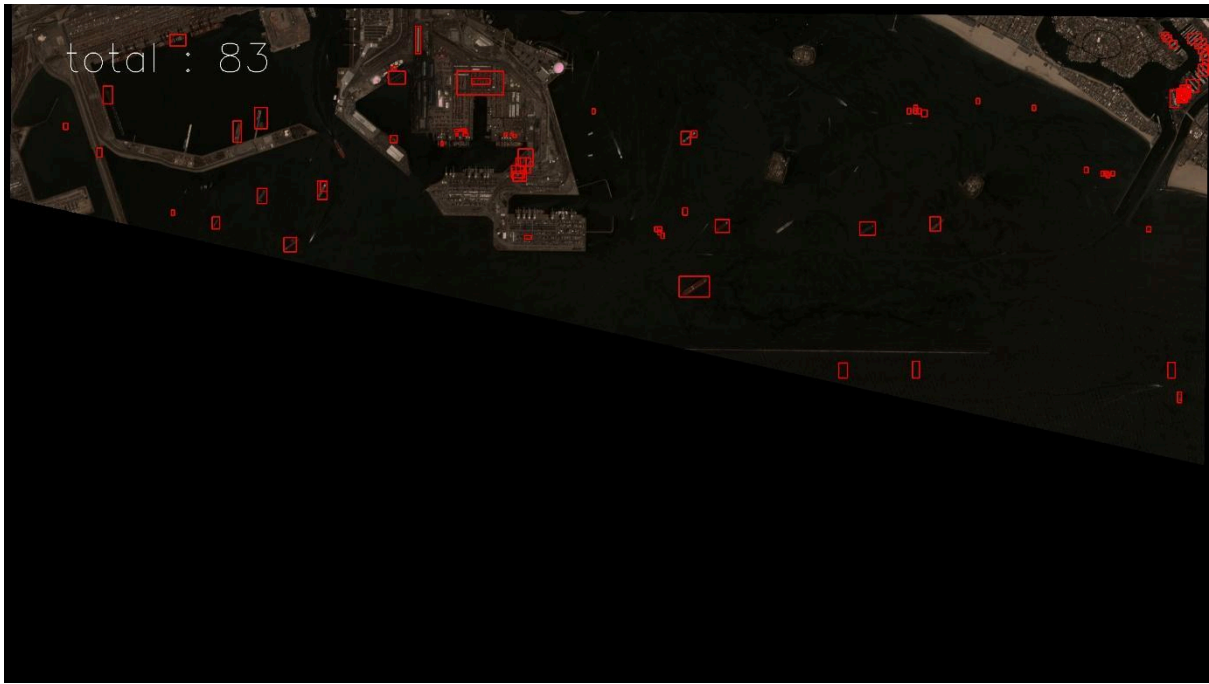
These trends were used as socio-economic validation for visual and satellite-based findings.

IV. Results and Observations

A. Correlation Across Data Streams

- **Satellite Detection Spikes** in August 2021 (e.g., 139 detections on Aug 2) closely aligned with known vessel backlogs.





Combined 139 ships were detected (56+83)



Aug 2 (2023)

- **Port Metrics** confirmed congestion, with flattened seasonal trends and prolonged turnaround times.
- **Media Trends** reflected public and industrial awareness, peaking during operational bottlenecks.

This triangulation confirms that remote sensing, operational data, and media signals offer mutually reinforcing perspectives.

B. Satellite Image Detection: 2021 vs. 2023

- **2021** imagery showed higher object detection counts on average compared to 2023.
- Cloud cover occasionally disrupted detection (handled by filtering dates with full cloud obscuration).

Date	2023	2021
01-Aug	36	48

02-Aug	no image or complete cloud cover	139
03-Aug	46	21
04-Aug	no image or complete cloud cover	60
05-Aug	no image or complete cloud cover	69
06-Aug	no image or complete cloud cover	0
07-Aug	no image or complete cloud cover	34
08-Aug	72	2
09-Aug	no image or complete cloud cover	52
10-Aug	no image or complete cloud cover	69
11-Aug	no image or complete cloud cover	16
12-Aug	no image or complete cloud cover	43
13-Aug	88	24
14-Aug	no image or complete cloud cover	107
15-Aug	51	120
16-Aug	no image or complete cloud cover	13
17-Aug	51	no_image
18-Aug	no image or complete cloud cover	no_image
19-Aug	no image or complete cloud cover	no_image
20-Aug	no image or complete cloud cover	12
21-Aug	no image or complete cloud cover	2
22-Aug	85	no_image
23-Aug	29	no_image
24-Aug	no image or complete cloud cover	94
25-Aug	48	no_image
26-Aug	no image or complete cloud cover	126
27-Aug	no image or complete cloud cover	73
28-Aug	no image or complete cloud cover	no_image

29-Aug	77	52
30-Aug	48	31
31-Aug	no image or complete cloud cover	no_image

V. Conclusion

This research demonstrates the viability of integrating satellite imagery, port analytics, and media trends to monitor and assess supply chain health. By applying lightweight CNNs to high-resolution satellite data and corroborating with vessel and news information, we can generate a composite picture of port congestion and logistical stress.

The results from the Port of Los Angeles during the 2021 crisis period validate this methodology. Future work could include real-time integration and ML-based forecasting models to automate congestion prediction globally.